

# Fine-Tuning HuggingFace Embedding Models for Asymmetric Search

Train embedders on your data

# Asymmetric Search: Why It Matters

- Short Queries  $\neq$  Long Chunks

## QUERY

What days of the week are campus tours offered at Christopher Newport University?

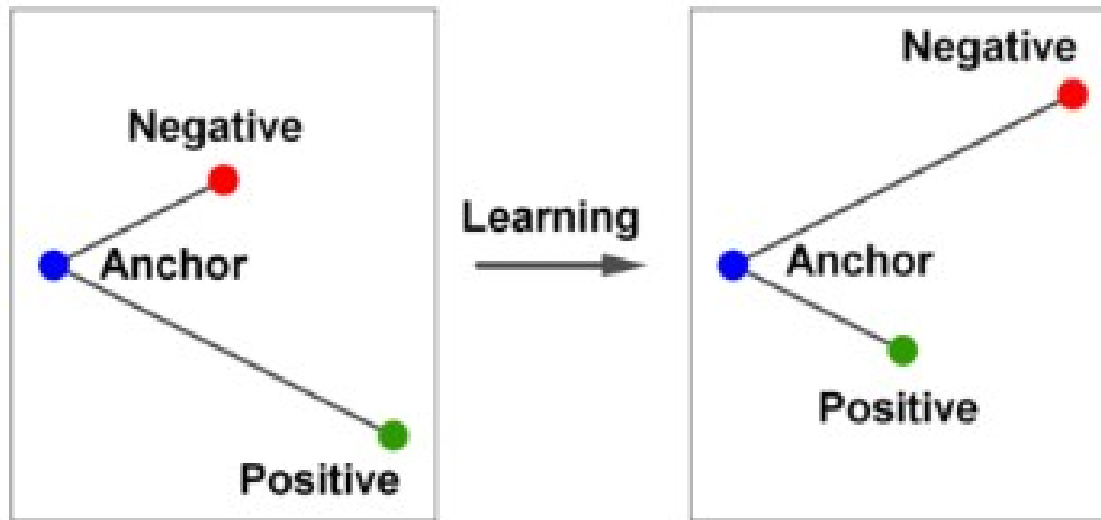
## Chunk

'Come See for Yourself \n' 'Ready to picture your life as a CNU Captain? Schedule \n' 'a visit and see why so many students choose to call Chris-\n' 'topher Newport home. \n' 'Campus tours are offered Monday through Saturday \n' '(except major holidays). Visit us online at admission.cnu. \n' 'edu or call (757) 594-7015/(800) 333-4268. \n' 'We look forward to showing you why Christopher New-\n' 'port is the number one regional public university in Virginia!'

- Embedder Model must learn directional alignment
- Goal: Fine tuning improves retrieval accuracy

# Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



Want to push negative away and pull positive closer

# Loss Functions

- MultipleNegativesRankingLoss (MNRL)
- Uses 'in-batch' negatives
- ex. For this batch of 4

|                                                   |                                                   |
|---------------------------------------------------|---------------------------------------------------|
| What days of the week are campus tours offered... | Come See for Yourself \nReady to picture your ... |
| Which minor subjects are offered under the cat... | Health Studies .....                              |
| Which minors require 143 credits?                 | Film Studies, minor .....                         |
| Which minor fields of study are offered in the... | Philosophy of Law, minor .....                    |

# Loss Functions

- MultipleNegativesRankingLoss (MNRL)
- Uses 'in-batch' negatives
- ex. For this batch of 4, **we get 3 NEGATIVES**

|                                                   |                                   |                 |                  |            |         |
|---------------------------------------------------|-----------------------------------|-----------------|------------------|------------|---------|
| What days                                         | question (or anchor)              | ours offered... | Come See for You | POSITIVE   | our ... |
| Which minor subjects are offered under the cat... |                                   |                 | He               | NEGATIVE 1 | .....   |
|                                                   | Which minors require 143 credits? |                 | Film             | NEGATIVE 2 | .....   |
| Which minor fields of study are offered in the... |                                   |                 | Philoso          | NEGATIVE 3 | .....   |

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|                                                   | Which minors require 143 credits? |                 | Film             | NEGATIVE 2 | .....   |
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So this row will expand into 4 rows

(Anchor, POSITIVE, Negative1), (Anchor, POSITIVE, Negative2) etc

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|                                                   |                                   |                 |                  |          |         |
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|                                                   | Which minors require 143 credits? |                 | Film             | NEGATIVE | .....   |
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The larger the batch the more negatives. But they are randomly picked, so easy negatives 😞

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|                                                   |                                   |                 |                  |          |         |
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We want hard negatives, those that are 'close' to the positive!



# Why Hard Negatives?

Not all negatives are equal

## Easy Negative

“How to request PTO?”

vs.

“Company stock plan details.”

**Too different**

(Likely what MNRL sees)

## Hard Negative

“How can I change my benefits?”

vs.

“What if I need to update my spouse’s benefits?”

**Semantically close, teaches model what NOT to retrieve**

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## Hard negatives improve:

- Recall@K
- MRR
- Generalization
- Ranking quality

# Loss Functions

## TripletLoss

- Uses explicit (anchor, positive, **negative**)
- Teaches model to distinguish between positive passages and confusing look-alikes

## Typical outcome

Big improvements in Recall@K

Sharper ranking between similar passages

# Loss Functions

TripletLoss

Uses explicit (anchor, positive, negative)

How does it work?

# Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



Margin is how much further out from positive that we will tolerate

# Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



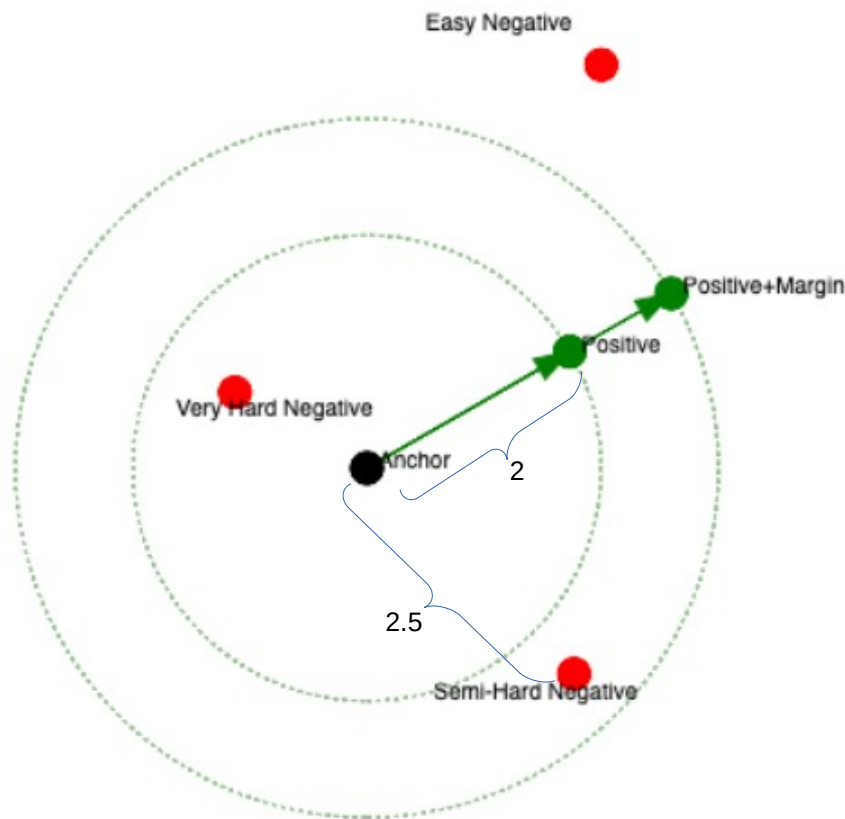
Margin=1

Loss= $\max(2-4+1, 0) \Rightarrow 0$

no loss, so nothing to learn here

# Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



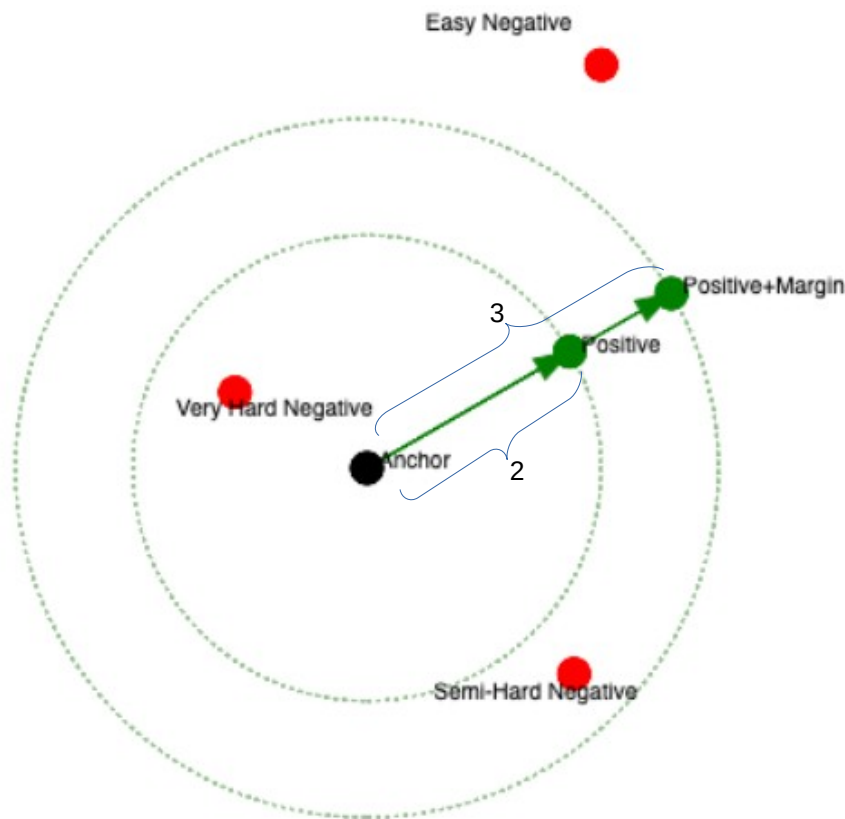
Margin=1

Loss= $\max(2-2.5+1, 0) \Rightarrow .5$

.5 loss learn from this one!

# Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



Margin=1

Loss= $\max(2-3+1, 0) \Rightarrow 0$  no learning from this one



# Loss Functions

$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



- Want 'hard' negatives
- Not 'easy' negatives
- Want negatives that are 'close'

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- **How?**

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$$\mathcal{L} = \max(d(a, p) - d(a, n) + \text{margin}, 0)$$



- Want 'hard' negatives
- Not 'easy' negatives
- Want negatives that are 'close'
- How?
- **Pull from Vector DB.**

# Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns

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|                 |               |
|-----------------|---------------|
| Original Anchor | Orig Positive |
|-----------------|---------------|

← Start with this

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- Encode all chunks → store in ChromaDB
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- Filter true positive → keep top 3–5 as Hns

|                 |               |                 |
|-----------------|---------------|-----------------|
| Original Anchor | Orig Positive | Hard Negative 1 |
| Original Anchor | Orig Positive | Hard Negative 2 |
| Original Anchor | Orig Positive | Hard Negative 3 |
| Original Anchor | Orig Positive | Hard Negative 4 |

Finish  
with this

# Hard Negative (HN) Mining

- Encode all chunks → store in ChromaDB
- Encode question → query top-K closest chunks
- Filter true positive → keep top 3–5 as Hns
- **This dramatically expands training dataset**

|                 |               |                 |            |
|-----------------|---------------|-----------------|------------|
| Original Anchor | Orig Positive | Hard Negative 1 | } New rows |
| Original Anchor | Orig Positive | Hard Negative 2 |            |
| Original Anchor | Orig Positive | Hard Negative 3 |            |
| Original Anchor | Orig Positive | Hard Negative 4 |            |

New Column

# Fine-Tuning

Better performance on domain-specific vocabulary (legal, medical, etc.)

- Improves retrieval precision/recall
- Reduces hallucinations in downstream RAG pipelines



# 2-Stage Embedder Fine-Tuning

- Stage 1: **Fine tune** model using MNRL on (question, chunk) pairs
- Embed corpus using fine tuned model
- Hard Negative Mine to expand training dataset
- Stage 2: **Fine tune** model using TripletLoss on (question, positive, hard negative)

# Key Takeaways

- MNRL uses easy negatives but will probably improve model
- Triplet loss uses hard negatives and will probably dramatically improve model
- Multi-stage finetuning is generally very effective, especially for specialized domains (medicine, law, etc)